

# Day 13: Area of a Triangle

LESSON + PRACTICE

MathCraft CPA · Chandler Park Academy · Grade 6 Mathematics · Professor Xavier Honablue, M.Ed.

**Standard:** 6.G.A.1 — find the area of right, acute, and obtuse triangles by composing into rectangles or decomposing into triangles; MP.3, MP.6

**I can:** find the area of any triangle with  $A = \frac{1}{2} \times \text{base} \times \text{height}$ , pick the height that is perpendicular to the base I chose, and run the formula backwards to find a missing base or height.

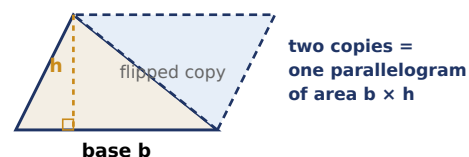
**Homework:** Earn 90% on IXL "Area of triangles" (GG, 6th grade) — [ixl.com/math/grade-6/area-of-triangles](https://www.ixl.com/math/grade-6/area-of-triangles)

**Remember:** show ALL work by hand in your math notebook (graph paper is best) and label every answer with units.

## Learn

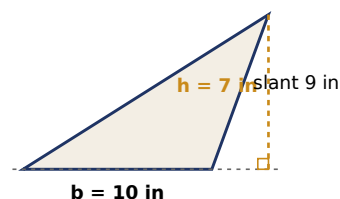
### 1 · Two triangles make a parallelogram

Take any triangle. Make an exact copy, flip the copy upside down, and slide it against the first one. The two pieces always lock together into a **parallelogram** with the **same base** and the **same height**. That parallelogram's area is  $b \times h$  (Day 8), and our triangle is exactly **half** of it. So for every triangle:  $A = \frac{1}{2} \times b \times h$ , which you can also read as "multiply the base by the height, then cut it in half."



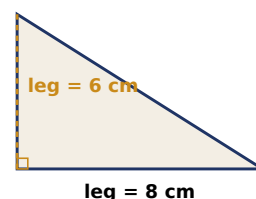
### 2 · The height is perpendicular — and it may sit outside the triangle

The height is the straight-up distance from the base to the opposite corner, meeting the base at a **right angle**. On a wide (obtuse) triangle, that straight-up line lands **outside** the triangle, so the base is extended with a dotted line. The slanted side is never the height — it is longer, and using it makes the area too big.



### 3 · In a right triangle, the two legs are the base and the height

A right triangle already has two sides that meet at  $90^\circ$ . Either leg can be the base; the other leg is then the height. The longest side (the one across from the right angle) is never used in the area formula.



### 4 · Same formula, run backwards

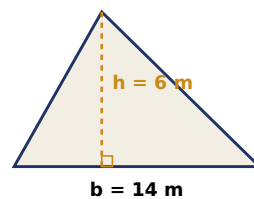
Because  $A = \frac{1}{2}bh$ , doubling the area undoes the half:  $2A = b \times h$ . So a missing base is  $b = 2A \div h$  and a missing height is  $h = 2A \div b$ . Always double first — forgetting to double is the most common error on this lesson.

## Worked Examples

**Example 1.** Find the area of a triangle with base 14 m and height 6 m.

- Step 1:  $A = \frac{1}{2} \times b \times h = \frac{1}{2} \times 14 \times 6$ .
- Step 2: Multiply the base and height first:  $14 \times 6 = 84$ .
- Step 3: Take half:  $84 \div 2 = 42$ .
- Step 4: Units:  $m \times m = \text{square meters}$ .

✓ **Area = 42 sq m**



**Example 2.** An obtuse triangle has base 10 in, height 7 in, and a slanted side of 9 in. Find the area.

- Step 1: Circle the base (10 in) and the height (7 in). The 9 in slanted side is a distractor.
- Step 2:  $A = \frac{1}{2} \times 10 \times 7 = \frac{1}{2} \times 70$ .
- Step 3:  $70 \div 2 = 35$ .
- Step 4: The height landed outside the triangle, but the formula does not change.

✓ **Area = 35 sq in**

**Example 3.** A triangle has an area of 54 sq ft and a base of 12 ft. Find its height.

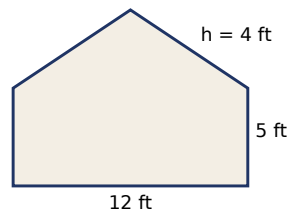
- Step 1: Start from  $A = \frac{1}{2} \times b \times h$ , so  $54 = \frac{1}{2} \times 12 \times h$ .
- Step 2: Double both sides to undo the half:  $108 = 12 \times h$ .
- Step 3:  $h = 108 \div 12 = 9$ .
- Step 4: Check:  $\frac{1}{2} \times 12 \times 9 = 54$ . ✓

✓ **Height = 9 ft**

**Example 4.** A wall is shaped like a rectangle 12 ft wide and 5 ft tall with a triangular top 12 ft wide and 4 ft tall. Find the total area.

- Step 1: Decompose: one rectangle + one triangle.
- Step 2: Rectangle:  $12 \times 5 = 60$  sq ft.
- Step 3: Triangle:  $\frac{1}{2} \times 12 \times 4 = 24$  sq ft.
- Step 4: Add:  $60 + 24 = 84$  sq ft.

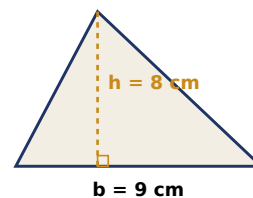
✓ **Area = 84 sq ft**



## Practice — show your work in your notebook

1. Find the area of the triangle.

Answer: \_\_\_\_\_



2. A right triangle has legs of 7 in and 10 in. Find its area.

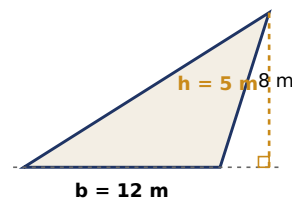
Answer: \_\_\_\_\_

3. Find the area of a triangle with base 15 ft and height 6 ft.

Answer: \_\_\_\_\_

4. **Distractor check.** This triangle is labeled base 12 m, height 5 m, and slanted side 8 m. Find its area.

Answer: \_\_\_\_\_



5. A triangle has an area of 48 sq cm and a height of 8 cm. Find its base.

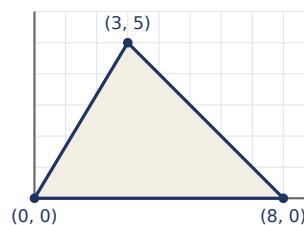
Answer: \_\_\_\_\_

6. A triangle has an area of 30 sq in and a base of 10 in. Find its height.

Answer: \_\_\_\_\_

7. Find the area of the triangle plotted on the grid.

Answer: \_\_\_\_\_



8. **Find the mistake.** For a triangle with base 16 ft and height 9 ft, Andre wrote  $A = 16 \times 9 = 144 \text{ sq ft}$ . What did he forget? What is the correct area?

Answer: \_\_\_\_\_

9. A doghouse wall is a rectangle 10 ft by 6 ft with a triangular roof 10 ft wide and 4 ft tall. Find the total area.

Answer: \_\_\_\_\_

10. A triangular garden has a base of 18 ft and a height of 7 ft. One bag of grass seed covers 30 sq ft. How many whole bags are needed?

Answer: \_\_\_\_\_

11. Find the area of a triangle with base 7.5 cm and height 4 cm.

Answer: \_\_\_\_\_

**12.** Find the area of a triangle with base 5 in and height  $3\frac{1}{2}$  in.

Answer: \_\_\_\_\_

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# Answer Key & Worked Solutions

Day 13 · Area of a Triangle · Teacher / family reference

**1.** Find the area of the triangle.

- Step 1:  $A = \frac{1}{2} \times b \times h = \frac{1}{2} \times 9 \times 8$ .
- Step 2:  $9 \times 8 = 72$ .
- Step 3:  $72 \div 2 = 36$ .

✓ **Answer: 36 sq cm**

**2.** A right triangle has legs of 7 in and 10 in. Find its area.

- Step 1: In a right triangle the two legs are the base and the height.
- Step 2:  $A = \frac{1}{2} \times 7 \times 10 = \frac{1}{2} \times 70 = 35$ .

✓ **Answer: 35 sq in**

**3.** Find the area of a triangle with base 15 ft and height 6 ft.

- Step 1:  $A = \frac{1}{2} \times 15 \times 6$ .
- Step 2:  $15 \times 6 = 90$ .
- Step 3:  $90 \div 2 = 45$ .

✓ **Answer: 45 sq ft**

**4. Distractor check.** This triangle is labeled base 12 m, height 5 m, and slanted side 8 m. Find its area.

- Step 1: Use only the base and the perpendicular height: 12 and 5.
- Step 2:  $A = \frac{1}{2} \times 12 \times 5 = \frac{1}{2} \times 60 = 30$ .
- Step 3: The 8 m slanted side is not used ( $12 \times 8 \div 2 = 48$  is the trap answer).

✓ **Answer: 30 sq m**

**5.** A triangle has an area of 48 sq cm and a height of 8 cm. Find its base.

- Step 1:  $48 = \frac{1}{2} \times b \times 8$ .
- Step 2: Double the area first:  $96 = b \times 8$ .
- Step 3:  $b = 96 \div 8 = 12$ .

✓ **Answer: 12 cm**

**6.** A triangle has an area of 30 sq in and a base of 10 in. Find its height.

- Step 1: Double the area:  $60 = 10 \times h$ .
- Step 2:  $h = 60 \div 10 = 6$ .

✓ **Answer: 6 in**

**7.** Find the area of the triangle plotted on the grid.

- Step 1: Base runs from (0, 0) to (8, 0): 8 units.
- Step 2: Height is the straight-up distance to  $y = 5$ : 5 units.
- Step 3:  $A = \frac{1}{2} \times 8 \times 5 = 20$ .

✓ **Answer: 20 square units**

**8. Find the mistake.** For a triangle with base 16 ft and height 9 ft, Andre wrote  $A = 16 \times 9 = 144$  sq ft. What did he forget? What is the correct area?

- Step 1: Andre found the area of the whole parallelogram, not the triangle.
- Step 2: A triangle is half of it:  $144 \div 2 = 72$ .
- Step 3: He forgot the  $\frac{1}{2}$ .

✓ **Answer: He left out the  $\frac{1}{2}$ ; correct area is 72 sq ft**

**9.** A doghouse wall is a rectangle 10 ft by 6 ft with a triangular roof 10 ft wide and 4 ft tall. Find the total area.

- Step 1: Rectangle:  $10 \times 6 = 60$ .
- Step 2: Triangle:  $\frac{1}{2} \times 10 \times 4 = 20$ .
- Step 3:  $60 + 20 = 80$ .

✓ **Answer: 80 sq ft**

**10.** A triangular garden has a base of 18 ft and a height of 7 ft. One bag of grass seed covers 30 sq ft. How many whole bags are needed?

- Step 1: Area:  $\frac{1}{2} \times 18 \times 7 = \frac{1}{2} \times 126 = 63$  sq ft.
- Step 2:  $63 \div 30 = 2.1$  bags.
- Step 3: You cannot buy part of a bag, so round up to 3.

✓ **Answer: 3 bags (63 sq ft to cover)**

**11.** Find the area of a triangle with base 7.5 cm and height 4 cm.

- Step 1:  $A = \frac{1}{2} \times 7.5 \times 4$ .
- Step 2:  $7.5 \times 4 = 30$  ( $75 \times 4 = 300$ , one decimal place).
- Step 3:  $30 \div 2 = 15$ .

✓ **Answer: 15 sq cm**

**12.** Find the area of a triangle with base 5 in and height  $3\frac{1}{2}$  in.

- Step 1:  $A = \frac{1}{2} \times 5 \times 3\frac{1}{2}$ .
- Step 2:  $5 \times 3\frac{1}{2} = 5 \times 3 + 5 \times \frac{1}{2} = 15 + 2\frac{1}{2} = 17\frac{1}{2}$ .
- Step 3: Half of  $17\frac{1}{2} = 8\frac{3}{4}$ .

✓ **Answer:  $8\frac{3}{4}$  sq in**